

### 4-3 Motor Start and Operation



- ☑ If the motor does not start within 1–2 seconds, turn off the power immediately. The motor must be thoroughly inspected and troubleshooted before it can be started again.
- ☑ Startup Time and Abnormal Noise: The startup time will increase for large load machines. However, for general machine startup, if the time is particularly long or difficult to complete the startup, or if there is an abnormally loud noise emitted during the startup, do not operate the motor and contact the original manufacturer as soon as possible.

After the motor has been installed, you need to run it with no load, and only after it is running normally you can connect it to a loaded machinery. Except under special circumstances, the motor is usually started with light load and slowly loaded at full speed for full load operation. When the deviation of the power supply voltage and frequency from the rated voltage and frequency values on the nameplate exceeds  $\pm 5\%$  and  $\pm 1\%$ , respectively, the motor cannot guarantee continuous output of the rated power, and continuous overloading is not allowed.

If you need to start and restart the motor too frequently, pay more attention on the input current limit (refer to Section 1-6 Motor Characteristic Curves). You can restart the motor when the startup fails, but you must pay attention to the over-current protection level.

The possible errors and corresponded troubleshooting are as follows:

| Fault Description                                                             | Possible Cause                                                                                                                                                                                                                                                                 | Corrective Actions                                                                                                                                                                                                                                                                                                                                                                                           |
|-------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Unable to activate                                                            | <ul style="list-style-type: none"> <li>● No power connected</li> <li>● Excessive load or stuck transmission</li> <li>● Control device wired incorrectly</li> <li>● Stator winding fault</li> </ul>                                                                             | <ul style="list-style-type: none"> <li>● Verify if the switches, fuses, contacts and wires function properly.</li> <li>● Choose a motor with larger capacity or reduce the load. Verify if the transmission mechanism is jammed and remove it asap.</li> <li>● Verify if the wiring is correct.</li> <li>● Verify if there's winding disconnection, short-circuit, grounding fault.</li> </ul>               |
| The speed of the motor is slower than the rated speed when running with load. | <ul style="list-style-type: none"> <li>● Motor is overloaded</li> <li>● Power supply voltage is too low</li> </ul>                                                                                                                                                             | <ul style="list-style-type: none"> <li>● Choose a motor with larger capacity or reduce the load.</li> <li>● Use a multi-meter to verify if the motor input supply voltage is normal.</li> </ul>                                                                                                                                                                                                              |
| Excessive motor temperature or smoke                                          | <ul style="list-style-type: none"> <li>● Motor is overloaded.</li> <li>● Motor runs without phase</li> <li>● Motor air duct is blocked</li> <li>● Ambient temperature rises</li> <li>● Stator winding fault.</li> <li>● Power supply voltage is too low or too high</li> </ul> | <ul style="list-style-type: none"> <li>● Choose a motor with larger capacity or reduce the load</li> <li>● Verify and troubleshoot the fuse, switch contact point. Clean the foreign object and dusts in the motor air duct.</li> <li>● Take cooling measure</li> <li>● Verify and repair the Stator winding.</li> <li>● Use a multi-meter to verify if the motor input supply voltage is normal.</li> </ul> |